

Birgenair flight 301

Flight model: Boeing 757

Date: February 6, 1996

From: Puerto Plata, Dominican Republic

To: Frankfurt, Germany

In this case the first officer, is new to the plane with few miles of flying in the same where as the captain has many hours of flying such planes.

Problem

Right at the time of take off, it is found that the captain's air speed instrument is not working. At 151, the plane hits **V1**; the point of no return. Later, the captain's airspeed indicator comes to life. After takeoff, auto pilot takes over. Minutes later the plane flashes 2 warnings stating that the plane is climbing too fast. But, the first officer's air speed indicator shows that it is climbing at normal speed. Later, the first officer's air speed indicator shows that the plane is traveling slowly and captain's indicator show the plane is traveling too fast. When the alarm is ignored, the over speed warning pops up. The captain slows down the plane. Then, the **stick shaker warning** pops up, which indicates a **stall** which occurs when the plane is going to attain a speed below which it cannot fly. In the **Altitude deviation indicator**, it represents more blue than brown which means the plane is in steep climb. Then, the plane enters in a steep drop. then, the thrust levers are set to full. The plane starts spiraling towards the sea.

Investigation

The air speed of the plane is sensed using a **pitot tube** which are located at the side of the nose of the plane. A faulty reading is reported when the pitot tube is blocked. When the plane rises in altitude the air blocked in the pitot tube would have expanded causing the air speed of the captain to increase after take off. The auto pilot gets the air speed information from the pilot's side of the plane. The faulty reading directed the auto pilot to take wrong decisions which made the nose to pitch up to slow it down which creates drag. The auto pilot is directed to pitch the nose not more than 15 degrees. When the over speed warning popped up in reality the plane was running at low speed. When the plane was running in low speed, when the thrust levers were set low, the plane started to drop. The auto pilot disconnects when the stick shakers warning is triggered. At one point, the crew did attempt to full engine thrust power, but the angle at which it was falling couldn't get the engine enough air. This made engine 1 to turn off which made it to turn towards one side. Before take off, when the 2 pilots instruments were indicating different reading, the take off should have been aborted.

Changes that were brought after the incident

In the pilots training, they must be subjected to a simulation test with block pitot tube scenario. The Boeing was also instructed to pop up warning when both the instruments on either side are contradicting. It had also allowed pilots to program which air speed reading to use when the flight is operated in auto pilot mode.

Reference

1. Fatal Alarms - The Mystery of Birgenair Flight 301 — Mayday: Air Disaster <https://www.youtube.com/watch?v=J1PwJb0kDmM>